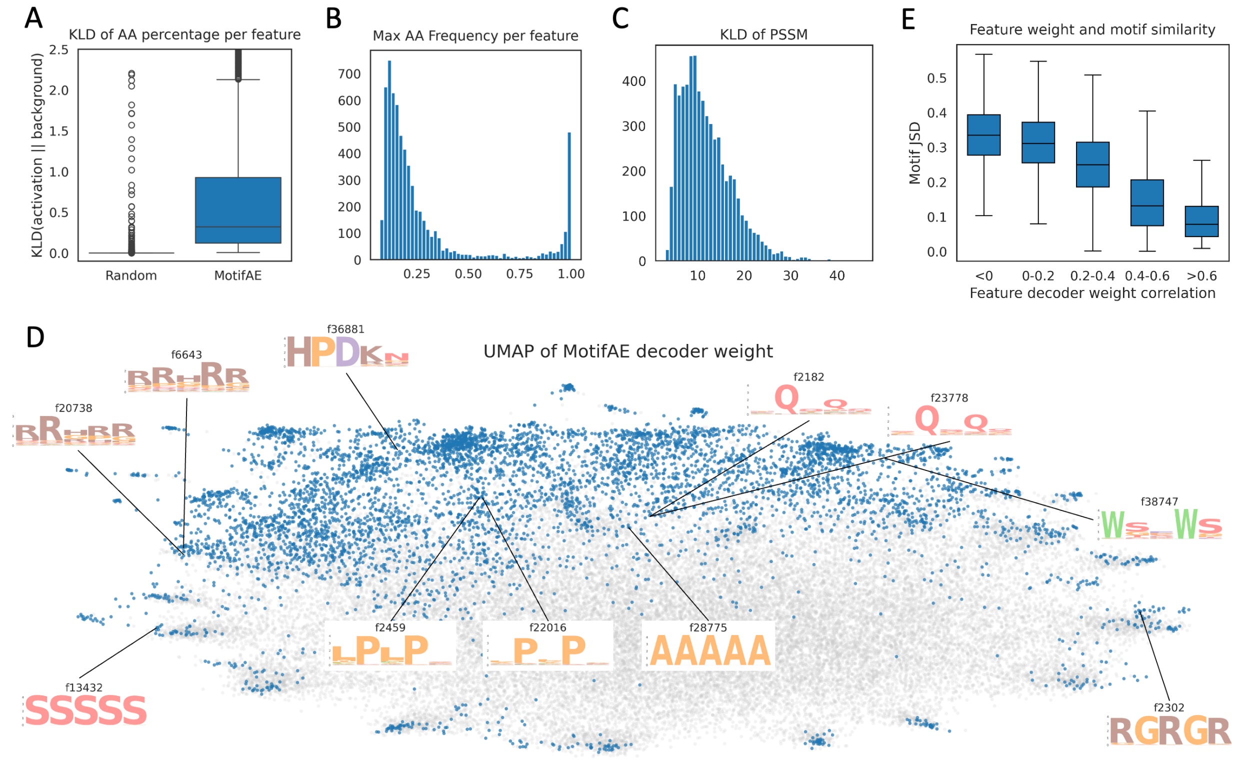

Figure 5. The universe of sequence patterns of MotifAE features.

(**A**) KL divergence between the amino acid composition of activated peptides of each feature and the background (amino acid composition in the 2.3M representative proteins). Random peptides matched in number and length distribution were used as a control. (**B**) Frequency of the most abundant amino acid in the activated peptides of each feature. (**C**) KL divergence of PSSMs calculated for each feature compared to the background amino acid frequency. KL divergence values were summed across the five aligned core positions. (**D**) UMAP visualization of MotifAE decoder weights. Each point represents a feature; gray points correspond to features without coherent activations, while blue points correspond to features with coherent activations. PSSM sequence logos are shown for some features. (**E**) Relationship between feature similarity in decoder weight space and similarity of their PSSMs. Decoder weight similarity was measured using Pearson correlation, while PSSM similarity was measured using Jensen–Shannon divergence (JSD), where smaller JSD values indicate higher similarity.

Aligning MotifAE with experimental data identifies features associated with the specific property.

Interpreting latent features remains a major challenge. While comparing MotifAE feature activations with known functional sites reveals the biological signal captured by latent features, this approach is limited by the scarcity of known functional sites. To overcome this, we developed a supervised approach to annotate MotifAE features using experimental data, which do not necessarily provide explicit site-level annotations. This approach involves two steps. First, a model is trained to predict experimental measurements from embeddings, many existing fine-tuning models can be directly reused; in this step, MotifAE reconstructions serve as a drop-in replacement for ESM2 embeddings. Second, a gating layer is introduced to selectively amplify or attenuate the activation magnitudes of MotifAE features. The gating layer is further trained on the same dataset, thereby modulating both the reconstructed embeddings and downstream predictions. After training, the gate weights reflect the association between features and the measured property, and the activated residues of associated features may underlie the property. We refer to this framework as MotifAE-G.

Here, we applied MotifAE-G to DMS experiments (**Figure 6A**). Because the ESM2 MLP layer's predicted amino acid probabilities correlate well with mutation effects (**Figure 2C-D**), we don't need to train an additional predictor (step one described above). A binary gate was used, meaning that each feature was either retained or completely suppressed. A differentiable Spearman correlation loss was used to update the gate parameters (see Methods). We used the mega-scale protein stability DMS dataset[32], which contains 379,495 single substitution mutations across 412 proteins, including both natural and designed proteins. On this dataset, MotifAE without gating performs comparably to ESM2 (**Figure S3A**). 412 proteins were grouped into 189 clusters based on 30% sequence identity. Among these, 285 proteins from 133 clusters were used for training and validation, while 127 proteins from the remaining 56 clusters were reserved for testing. Using the MotifAE-G framework, we identified 1,404 stability-associated features from the training set. With these features, MotifAE-G achieves a median Spearman correlation of 0.61 on the training set, significantly outperforming ESM2 (0.43), with 97.5% of training proteins showing performance improvement (**Figure 6B**). On the test set, MotifAE-G achieves a median Spearman correlation of 0.61, also significantly higher than ESM2 (0.44), with 88% of test proteins showing improvement (**Figure 6C**). For both training and test sets, the improvement was more pronounced for proteins that were initially predicted poorly by ESM2 (**Figure 6B-C**).

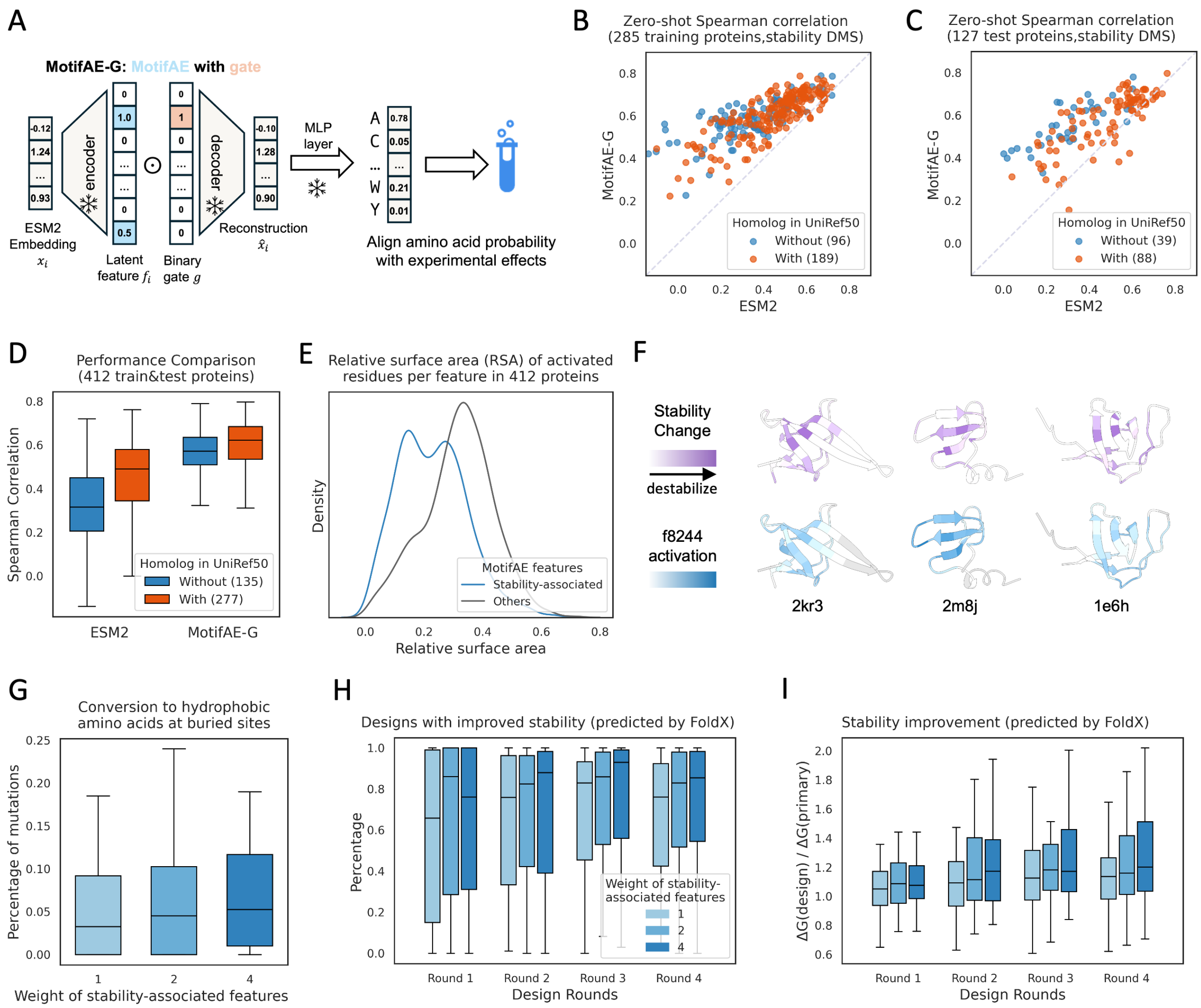

Figure 6. Aligning MotifAE with experimental data for feature annotation and protein design.

(**A**) Schematic of applying MotifAE-G framework to DMS data. A learnable binary gate was used to select latent features to align predicted amino acid probability with experimental mutation effect data; during training, only the gate parameters were updated. (**B-C**) Comparison of MotifAE-G and ESM2 performance on stability mutation effects across training and test proteins; each point represents a protein; color represents whether the protein has homolog in UniRef50. (**D**) Comparison of MotifAE-G and ESM2 performance on protein with and without homolog in UniRef50. (**E**) Mean relative solvent accessibility (RSA) of activated residues for stability-associated versus other MotifAE features, only features with at least 50 activated residues in 412 proteins were analyzed. (**F**) AlphaFold2-predicted structures of representative proteins. Upper panels show experimentally measured stability changes, averaged over all mutations at each residue (purple indicates stronger destabilizing effects). In the lower panel, the saturation of blue indicates the relative activation values of the MotifAE feature. (**G–I**) Designing proteins with improved stability using iterative redesign sampling. Gate weights of 1, 2, and 4 were applied to stability-associated features (shown in progressively darker blue), while other features were fixed at a gate weight of 1. (**G**) Buried residues were defined as sites with RSA < 0.2. All mutations from four rounds of redesign were included in the analysis. (**H**) Percentage of designs showing improved stability compared to their primary (unmutated) sequences. (**I**) Relative FoldX ΔG compared to the primary sequence, 1 indicates no improvement. The median ΔG across all designs in each round for each protein is shown. **H-I** show results for 32 test proteins, only designs with ESMFold pLDDT > 0.8 and FoldX ΔG > 2 kcal/mol were included.

We previously observed that ESM2 models perform better on natural proteins but worse on designed proteins[33]. Among the 412 proteins in the dataset, 135 have no homologs in UniRef50[34] (the training set of ESM2), ESM2 performs much worse on these proteins (median Spearman correlation 0.32) compared to proteins with homologs (0.49; **Figure 6D**). MotifAE-G improves prediction performance for both, achieving median Spearman correlations of 0.57 for proteins without homologs and 0.62 for proteins with homologs (**Figure 6D**).

To further assess the biophysical relevance of the stability-associated features, we analyzed the relative solvent-accessible surface area of activated residues of each feature. We found that the stability-associated features tend to activate at more buried residues (**Figure 6E**), consistent with the fact that stability-disrupting mutations often occur at buried residues[24]. Furthermore, we compared the activation value of each feature with the mean mutation effect per residue within each protein. We found that stability-associated features tend to exhibit stronger correlations with mean mutation effects, indicating better alignment with experimentally observed stability changes (**Figure S3B**). One example feature is f8244, which shows strong correlations across multiple proteins, with representative cases visualized in **Figure 6F**: residues with high f8244 activation tend to harbor destabilizing mutations. Together, these results demonstrate that by aligning latent features with experimental data, MotifAE-G identifies meaningful features associated with specific functions or protein properties, which in turn enhance the prediction of mutational effects for both natural and designed proteins, highlighting its strong generalization.

Steering MotifAE-G enables designing protein with enhanced stability.

pLMs can be used for protein design by sampling from their predicted amino acid probabilities. However, directly generating sequences from pLMs does not necessarily yield proteins with the desired property. To address this, several approaches have been proposed to fine-tune pLMs to guide the design process toward specific functions or properties[35,36]. Here, we explored whether MotifAE-G can be used for property-specific protein design. Specifically, we steered the stability-associated features identified from the DMS dataset to design proteins with improved stability. Because ESM2 is not suitable for de novo sequence generation, we adopted an iterative redesign sampling strategy[37]. At each iteration, a single mutation was sampled according to the probabilities predicted by MotifAE-G, using higher gate weights to amplify the influence of stability-associated features and gate weights of one for other features (see Methods). This process was repeated over multiple rounds to progressively refine the sequence. The designed proteins were evaluated using FoldX[38], a physics-based force-field model that can estimate folding free energy (ΔG), applied to structures predicted by ESMFold[10] (see Method).

We applied the above strategy to the representative proteins from the 56 test protein clusters in the DMS dataset, among them, 32 proteins with ESMFold pLDDT > 0.8 and FoldX-predicted ΔG > 2 kcal/mol were selected for subsequent design. We applied weights of 1, 2, and 4 to the stability-associated features, with a weight of 1 as the baseline, equivalent to design using ESM2. For each weight, four rounds of iterative redesign were applied to each protein. We did not conduct more rounds because introducing a few new core interactions is typically sufficient to stabilize the protein, whereas excessive mutations may alter the native fold. The procedure was repeated 100 times per protein. By analyzing the introduced mutations, we found that increasing the weights of stability-associated features resulted in a higher proportion of mutations converting other amino acids into hydrophobic residues at buried sites (**Figure 6G**). This observation aligns with the fact that proteins are primarily stabilized by hydrophobic interactions within their cores. Furthermore, analysis of FoldX-predicted ΔG values showed that higher weights on stability-associated features generated more designs with improved stability relative to the original proteins (**Figure 6H**) and produced greater improvements in predicted ΔG on average across four design rounds (**Figure 6I**).

Discussion

In this study, we present MotifAE, an unsupervised framework for extracting interpretable functional motifs from pLMs. By incorporating a local similarity loss, MotifAE encourages coherent feature activation that reflects the sequential continuity of motifs and basic structural elements, thereby improving its ability to discover biologically meaningful motifs. Analysis of MotifAE feature sequence patterns reveals rich diversity, ranging from single amino acid specificity to multiple amino acids consensus motifs, highlighting its potential for large-scale motif discovery. Furthermore, MotifAE-G demonstrates how unsupervised MotifAE features can be aligned with experimental data, enabling the identification of features associated with specific functions or properties. Beyond interpretation, feature selection also allows MotifAE-G to enhance predictive performance on related tasks and to guide the rational design of proteins with desired characteristics.

While some SAEs have been trained for pLMs[19,21,39], they directly adopted the training and interpretation strategies developed for large language models. Our work introduces methodological advances in both model training and